# Time Teaches: "The Clock as a Mathematical Object"

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The hidden structure of numbers.
We are going to look at the clock.
Not at the time, but at the clock itself.

In this lecture, all positions start at 1,
not at 0. This is a deliberate choice,
and it fits how the clock itself works.
An hour hand points to 1,
not to 0.
The digital root runs from 1 to 9,
not from 0 to 8.
The month of January is month 1,
not month 0.

Concretely: the second runs from 1 to 60,
the minute from 1 to 60,
the hour from 1 to 24.
When a position is at its maximum,
it stays at that maximum (60, not 0).
That is exactly how the digital root works:
at 9 it stays 9, not 0.

## The basic example

It is 10 o'clock. You add 3 hours.
What time is it?

You can calculate it: 10 + 3 = 13.
But when you look at the clock, you do not see 13.
The hour hand points to 1. So you say:
it is 1 o'clock.

Not because 13 and 1 are the same.
But because the clock has only 12 positions.
When you go past 12, it starts again.
The position it points to, we call 1.

Yet you also know that it is 13 hours.
The clock only shows you the position.
The rest you do entirely yourself.

You write this down like this:

$13 \equiv 1 \pmod{12}$

You read this as:
13 points to the same position as 1
on a 12-hour clock.

But there is a catch.

When the clock points to position 1,
how do you know whether it is 1 o'clock at night
or 1 o'clock in the afternoon?

The answer: you cannot see that from the clock.
You look outside,

or you know whether it is day or night.
But the clock itself does not tell you.

What the clock does do
is make two full rounds per day.
One round during the day, one round at night.
That is the double cycle.
And this double cycle is what makes the difference between day and night possible.

Without that double cycle you would never know whether it is 01:00 or 13:00.
With that double cycle you do know.
The double cycle is what sets the clock apart from a simple counter.
A counter is not a spiral but linear.

## Position and quantity

The clock has 12 positions.
Those positions are numbered 1 to 12.
That is what you see when you look at the clock.

The time that has passed since the starting point is a quantity. It can be any number:
1 hour, 13 hours, 25 hours, 37 hours. No limit.

Some quantities point to the same position:

- 1 hour
- 13 hours
- 25 hours
- 37 hours

All these moments point to position 1.
What they have in common is not the number itself, but the distance between them.
That distance is 12.
And that is why we say: the modulus is 12.

We write:

$13 \equiv 1 \pmod{12}$
$25 \equiv 1 \pmod{12}$
$37 \equiv 1 \pmod{12}$

The modulus 12 is what connects all these numbers. How many full rounds of 12
are there in each of those numbers?

In 13 there is one.
In 25 there are two.
And two rounds of 12 hours is exactly one full day.
This is where the structure becomes visible.

## A new perspective – the main theory

The contribution is structural. This work offers a new perspective on the way we look at the clock. And on how it works.
What we investigate is the structure behind
the measurement of time.

That structure has a form:

$p \equiv 1 \pmod{q}$

That is the main theory.
Everything that follows is an elaboration of it.
The clock is one instance of that form.

## Two forms: theoretical and actual

The main theory has two forms.
One is theoretical,
the other is actual.

The main theory is: $p \equiv 1 \pmod{q}$.

The theoretical form is: $q + 1 = p$.
So $p \equiv 1 \pmod{q}$, with $m = 1$.
That is one round. You arrive at position 1.
You are one level higher.

The actual form is: $2q + 1 = p$.
So $p \equiv 1 \pmod{q}$, with $m = 2$.
That is two rounds.
You arrive at position 1.
You are two levels higher.

The modulus stays the same.
Only the number of rounds changes.

And it is a spiral.
You keep coming back to position 1,
but at a higher level.
The position is the same.
The level rises.

## The Process and the cycle

To understand the spiral properly,
we need to talk about the Process.
A process is the movement you make
to go from one level to the next.
In the actual form ($m = 2$), a process is always two rounds.

Let us take the week as an example:

- The process of the week: $15 \equiv 1 \pmod{7}$. This is one process (two rounds of 7). And this process yields one week, but with two different moments.
- The cycle: if you do this process twice, you get the cycle.

The unit here is the half day (12 hours),
just like the 12-hour clock face.
A round of 7 is then 7 half days.
That is half a week (84 hours).
Two rounds are 14 half days.
That is a whole week (168 hours).

Now we come to the month.
The calendar calls this "30 days".
But mathematically, 30 ($2 \times 15$) is only half of the process.
To get the real, complete cycle of the month,
the one that resonates with all the moments, we have to do the process once more:
$2 \times 30 = 60$.

So:

- 15 = 1 week (1 process, 2 moments).
- 30 = 2 weeks (2 processes, 4 moments). This is half of the month cycle.
- 60 = 4 weeks (4 processes, 8 moments). This is the complete month cycle.

Note: 15, 30 and 60 are numbers of half days here.
60 half days are 30 days.

Just as on the clock we do not say:
"An hour has passed" when only half an hour has passed.
Because 30 min is half an hour.
And 60 is 1 hour.
The whole calendar works this way.
Because 30 days is half of 60.
That does not mean 60 days.
Just as $25 \equiv 1 \pmod{12}$ does not consist of units of 25 or 24 hours. The unit we know is twelve, and in fact it is $12 \times 2$.
With two different moments.
And that double moment is $m = 2$.

So for the month we do not say "it is only 30", but we use 60 (4 weeks)
to show the complete cycle.

### Not a sum, but a spiral

It is not a sum, but a spiral to ever higher levels.
Every round ends at position 1 of the next level.
There is no month 0, no day 0 and no minute 0.

**The week.**
A week has 7 days, that is 14 half days (168 hours).
The 15th half day is position 1 of the next week.
That is $15 \equiv 1 \pmod{7}$, with $m = 2$.

**The month.**
A month of 30 days is 60 half days.
And $60 = 4 \times 15$.

- 4 weeks are $4 \times 14 = 56$ half days = 28 days.
- Four leftovers remain: the four times that a week ends at position 1 of the next one. That is 4 half days = 2 days.
- Together: $28 + 2 = 30$ days.

Those 2 days are therefore not added separately.
They are those four leftovers together.
After day 30 you are at day 1 of the next month.

**The hour.**
It is the same pattern.
15 minutes fits 4 times into 60 minutes.
Four quarters of an hour make one hour.

- $4 \times 14 = 56$ minutes, plus the four leftovers $(4 \times 1) = 60$.
- After that you are in the next hour, at minute 1: $61 \equiv 1 \pmod{60}$.

That is the formula $X = 8q + 4$ with $q = 7$.
The $8q = 56$ is four times two rounds of 7.
The +4 are the four leftovers.
Both the month (60 half days) and the hour (60 minutes) follow this pattern.

**The seasons.**
Four seasons of 3 months make 12 months.
Four times 90 days make 360 days.
After the fourth season you are at month 1 of the next year.

## The spiral in practice

Let us show the spiral and the processes across the whole chain. We do not count the separate rounds, but the levels at which we arrive.

Two rounds are always one whole unit.
One round is half of that.

**The clock:**

- 13 ≡ 1 (mod 12). One round. Half a day.
- 25 ≡ 1 (mod 12). Two rounds. A whole day.
- 37 ≡ 1 (mod 12). Three rounds. A day and a half.
- 49 ≡ 1 (mod 12). Four rounds. Two days.
- 61 ≡ 1 (mod 12). Five rounds. Two and a half days.

**The week:**

- 8 ≡ 1 (mod 7). One round. Level 1 (half a week, 84 hours).
- 15 ≡ 1 (mod 7). Two rounds (1 process). Level 2 (1 week, 168 hours, with two moments).
- 22 ≡ 1 (mod 7). Three rounds. Level 3 (a week and a half).
- 29 ≡ 1 (mod 7). Four rounds (2 processes). Level 4 (2 weeks).

**The digital root:**

- 10 ≡ 1 (mod 9). One round.
- 19 ≡ 1 (mod 9). Two rounds.
- 28 ≡ 1 (mod 9). Three rounds.
- 37 ≡ 1 (mod 9). Four rounds.

**The week as a number:**

- 4 ≡ 1 (mod 3). One round.
- 7 ≡ 1 (mod 3). Two rounds.
- 10 ≡ 1 (mod 3). Three rounds.
- 13 ≡ 1 (mod 3). Four rounds.

**The month:**

- 31 ≡ 1 (mod 30). One round: half a month.
- 61 ≡ 1 (mod 30). Two rounds (1 process): the whole month.
- 91 ≡ 1 (mod 30). Three rounds: a month and a half.

**The minute:**

- 61 ≡ 1 (mod 60). One round: the half hour.
- 121 ≡ 1 (mod 60). Two rounds: the whole hour.
- 181 ≡ 1 (mod 60). Three rounds: an hour and a half.

**The year (the double cycle of 6):**

Just as 12 is double 6,
the year (12 months) is the double cycle of half a year (6 months).

- The 6 is linked to the week, because 7 ≡ 1 (mod 6).
- The year is therefore a beautiful spiral of 2 × 6 months.

The position is always 1.
The level rises.
That is the spiral.

## The main theory in one list

- The main theory is: p ≡ 1 (mod q).
- The ratio is: m = (p – 1) / q.
- The theoretical form is: m = 1, so p = q + 1.
- The actual form is: m = 2, so p = 2q + 1.
- The parameter choice of the clock is: (p, q) = (25, 12).
- The m of the clock is: 2.

The clock is therefore not an isolated system.
It is a parameter choice within a general form p ≡ 1 (mod q).

## The clock as a parameter choice

The clock has a specific parameter choice:
p = 25 and q = 12.

The ratio that goes with it:
m = (25 – 1) / 12 = 2.

The bases of the clock together form
a chain.
But that chain falls apart into two parts:
time and days.

**The time series:**

The millisecond is the smallest unit.
If there is a smaller unit, then that is
the smallest. But here we take the millisecond as the smallest.

- The second is 1,000 milliseconds.
- The minute is 60 seconds.
- The hour is 60 minutes.

**The days series:**

- The day is 24 hours. (25 ≡ 1 (mod 12))
- The month is 30 days. (31 ≡ 1 (mod 30) is one round, half a month; 61 ≡ 1 (mod 30) is the actual form, the whole month.)
- The year is 12 months. (25 ≡ 1 (mod 12))
- The decade is 10 years. (10 ≡ 1 (mod 9))
- The century is 10 decades. (10 ≡ 1 (mod 9))
- The millennium is 10 centuries. (10 ≡ 1 (mod 9); the digital root works only with 1 to 9.)

**The week, separately:**

- The week is 7 days.

30 is not a multiple of 7.
That is why the week stands apart.
But the week is modularly linked
to the month and the rest of the chain.
We will come back to that later.

This chain determines the structure of the clock.
The cycle closes after 2,520,000 days,
which is exactly 7,000 years.

## What we are going to do

The clock is not only an instrument
for measuring time.
It is also a building made of numbers.
And we can take that building apart.

The questions:

- What is a clock, really?
- What kind of object is it?
- Which numbers does it use?
- And what does it tell us when we write it out?

We start with what everyone knows.
And we end with something almost nobody has noticed.
Everything is explained along the way.
No prior knowledge needed.

## Step 1 – The clock

Look at a clock with a second hand.
The second hand ticks.
Every tick is one second.
After 60 ticks something has happened:
the minute hand has moved one step.
Sixty seconds together make one minute.

The minute hand does the same
on a larger scale.
Every 60 minutes something has happened:
the hour hand has moved one step.
Sixty minutes together make one hour.

The hour hand does the same
on an even larger scale.
Every 24 hours something has happened:
a new day begins.
Twenty-four hours together make one day.

What you see here is that the clock is a system that divides time into fixed chunks.
Every chunk has a fixed size relative to the previous one.
Those numbers – 60, 60, 24 – were fixed thousands of years ago and have not changed much since.

## Step 2 – The numbers the clock uses

Now we look more broadly. The clock does not only use hours and minutes. There are also days, weeks, months, years, decades, centuries and millennia. Each with its own number.

From small to large:

- The millisecond is the smallest unit.
- The second is 1,000 milliseconds.
- The minute is 60 seconds.
- The hour is 60 minutes.
- The day is 24 hours.
- The week is 7 days.
- The month is 30 days.
- The year is 12 months.
- The decade is 10 years.
- The century is 10 decades.
- The millennium is 10 centuries.

Every number in this list is a base:
it is the number of the smaller unit
needed to make one of the larger.

What you see here:
the clock does not use one number.
It uses several:

1,000 60 60 24 7 30 12 10 10 10

When you put them side by side,
you see that they all belong to the same system. They mesh like the gears

of a clock.

## Step 3 – The clock as three lists

A moment in time consists of three parts:
the hour of the day, the date and the weekday.
Each part is a list of numbers,
and every number starts at 1.

**The hour of the day:**

- The hour runs from 1 to 24.
- The minute runs from 1 to 60.
- The second runs from 1 to 60.
- The millisecond runs from 1 to 1,000.

**The date:**

- The day of the month runs from 1 to 30.
- The month runs from 1 to 12.
- The year runs from 1 to 10.
- The decade runs from 1 to 10.
- The century runs from 1 to 10.
- The millennium keeps counting: 1, 2, 3, ... with no upper limit.

**The weekday:**

- The weekday runs from 1 to 7.

An example:
it is millennium 3, century 5, decade 2, year 4, month 7, day 15, and the time is 22 hours, 35 minutes, 40 seconds and 800 milliseconds.

You write that like this:

date: (3, 5, 2, 4, 7, 15)
time: (22, 35, 40, 800)

The weekday stands next to it separately.

The millennium therefore has no upper limit.
The series can be extended to 100,000 years, a million years or further.
That is why we speak of several bases.

The week cycle runs separately.
It does not hang in the chain of months and years, because a week does not fit into a month:
a month has 30 days
and 30 is not a multiple of 7.

What you see here: every position runs from 1
up to and including its upper limit.
The second stands at 60, not at 0,
and the month starts at 1.
The digital root works this way too.
It runs from 1 to 9.
That is why at a multiple of 9 it always stays
at 9.

This is a deliberate choice. In most systems a position runs from 0 up to base minus one. Here it runs from 1 up to the base.
That fits how we read the clock.
This is called a positional system.
The decimal system is one.
The clock is the same idea,

but with different bases
and with positions that start at 1.

## Step 4 – From the list back to one number

Every position has a weight:
how much that position is worth on its own.
The weight is the product of all the bases below it.

**Within the day, in milliseconds:**

- The millisecond weighs 1.
- The second weighs 1,000.
- The minute weighs 60,000.
- The hour weighs 3,600,000.
- The day weighs 86,400,000.

**In the date, in days:**

- The day weighs 1.
- The month weighs 30.
- The year weighs 360.
- The decade weighs 3,600.
- The century weighs 36,000.
- The millennium weighs 360,000.

A year here has 12 · 30 = 360 days
and a millennium 1,000 years, so 360,000 days. In this way every moment gets one number:
the number of days since the starting point,
and within that.

## Step 5 – From the number to the list

Take N = 100,000,000 milliseconds.
You divide by the weights, one by one.
What is left after each division we call
the remainder.

- The days: 100,000,000 ÷ 86,400,000 = 1. The remainder is 13,600,000.
- The hours: 13,600,000 ÷ 3,600,000 = 3. The remainder is 2,800,000.
- The minutes: 2,800,000 ÷ 60,000 = 46. The remainder is 40,000.
- The seconds: 40,000 ÷ 1,000 = 40. The remainder is 0.
- The milliseconds: 0 ÷ 1 = 0.

That is elapsed time: 1 day, 3 hours, 46 minutes,
40 seconds and 0 milliseconds.

A position on the clock counts from 1.
So you add 1 to each part:

elapsed: (1 day, 3 hours, 46 min, 40 s, 0 ms)
position: (in day 2, in hour 4, in minute 47, in second 41, in ms 1)

When the third hour is over,
the clock points to the fourth hour.
That is exactly how you read the clock.

What you see here:
the series of weights falls from 86,400,000 to 1. Every step divides by the base:
86,400,000 ÷ 24 = 3,600,000;
3,600,000 ÷ 60 = 60,000;

60,000 ÷ 60 = 1,000;
1,000 ÷ 1,000 = 1.
So the series of weights ends at 1,
the smallest unit.
And the series of remainders falls from 13,600,000 to 0. Every step leaves less: 13,600,000 → 2,800,000 → 40,000 → 0 → 0.
So the series of remainders ends at 0, no remainder. Both series meet at the +1:
position = the remainder + 1.

What you see here:
the clock falls apart neatly into pieces.
No remainder, no loss.
Every number belongs to exactly one list and every list to exactly one number.
That is called a bijection within the cycle.
And this bijection works in two directions:
from N to the list (counting down to 0) and from the list to N (adding up from 1).

## Step 5b – Three more examples, and the way back

**Example 2: N = 215,436,789 milliseconds**

- The days: 215,436,789 ÷ 86,400,000 = 2. The remainder is 42,636,789.
- The hours: 42,636,789 ÷ 3,600,000 = 11. The remainder is 3,036,789.
- The minutes: 3,036,789 ÷ 60,000 = 50. The remainder is 36,789.
- The seconds: 36,789 ÷ 1,000 = 36. The remainder is 789.
- The milliseconds: 789 ÷ 1 = 789. The remainder is 0.

elapsed: (2 days, 11 hours, 50 min, 36 s, 789 ms)
position: (in day 3, in hour 12, in minute 51, in second 37, in ms 790)

**Example 3: N = 0**

elapsed: (0, 0, 0, 0, 0)
position: (day 1, hour 1, minute 1, second 1, ms 1)

That is the starting point. Everything stands at 1.

**Example 4: N = 86,400,000 (exactly one whole day)**

elapsed: (1 day, 0 hours, 0 min, 0 s, 0 ms)
position: (day 2, hour 1, minute 1, second 1, ms 1)

**The way back:**

You multiply each elapsed part by its weight and add them up.

For example 1:

1 · 86,400,000 + 3 · 3,600,000 + 46 · 60,000 + 40 · 1,000 + 0 · 1
= 86,400,000 + 10,800,000 + 2,760,000 + 40,000
= 100,000,000

For example 2:

2 · 86,400,000 + 11 · 3,600,000 + 50 · 60,000 + 36 · 1,000 + 789
= 172,800,000 + 39,600,000 + 3,000,000 + 36,000 + 789
= 215,436,789

Both times you arrive back at the starting number N.
That is the bijection in two directions.

## Step 6 – The theoretical form: one round

Let us first go through the theoretical form
properly.

That is the basis.
Then we will bring out the actual form.

The main theory is: $p \equiv 1 \pmod{q}$.

The theoretical form is: $q + 1 = p$.

That means:
you go round once,
and you arrive at position 1.
You are one level higher.

Examples:

- The clock: $12 + 1 = 13$. And $13 \equiv 1 \pmod{12}$. That is one round of 12 hours. That is the half day.
- The week: $7 + 1 = 8$. And $8 \equiv 1 \pmod{7}$. That is one round of 7. That is the half week.
- The digital root: $9 + 1 = 10$. And $10 \equiv 1 \pmod{9}$. That is one round of 9.
- The week as a number: $3 + 1 = 4$. And $4 \equiv 1 \pmod{3}$. That is one round of 3.
- The minute: $60 + 1 = 61$. And $61 \equiv 1 \pmod{60}$. That is one round of 60. That is the half hour.
- The month: $30 + 1 = 31$. And $31 \equiv 1 \pmod{30}$. That is one round of 30. That is the half month.

This is the basis. One round. $m = 1$.

## Step 6b – The actual form: two rounds

Now we bring out the actual form.
That is the form the clock really uses.

The actual form is: $2q + 1 = p$.

That means: you go round twice,
and you arrive at position 1.
You are two levels higher.
Only now is a whole unit complete:
a whole day, a whole week.

Examples:

- The clock: $2 \times 12 + 1 = 25$. And $25 \equiv 1 \pmod{12}$. That is two rounds of 12 hours. And two rounds of 12 hours = 1 day (with a day moment and a night moment).
- The week: $2 \times 7 + 1 = 15$. And $15 \equiv 1 \pmod{7}$. That is two rounds of 7. And two rounds of 7 = 1 week (with two different moments).
- The digital root: $2 \times 9 + 1 = 19$. And $19 \equiv 1 \pmod{9}$. That is two rounds of 9.
- The week as a number: $2 \times 3 + 1 = 7$. And $7 \equiv 1 \pmod{3}$. That is two rounds of 3.
- The minute: $2 \times 60 + 1 = 121$. And $121 \equiv 1 \pmod{60}$. That is two rounds of 60. And two rounds of 60 = 1 hour (with two different moments).
- The month: $2 \times 30 + 1 = 61$. And $61 \equiv 1 \pmod{30}$. That is two rounds of 30. And two rounds of 30 = 1 month (with two different moments).

This is the actual form.
Two rounds. $m = 2$.

## Step 6c – The Process and the spiral

The spiral is everywhere.
And to understand it properly,
we look at the processes.
A process is the movement of the actual form ($m = 2$).

**The process of the week:**

- $8 \equiv 1 \pmod{7}$. One round. Level 1 (half a week).

- 15 ≡ 1 (mod 7). Two rounds (1 process). Level 2 (1 week with two moments).
- 22 ≡ 1 (mod 7). Three rounds. Level 3 (a week and a half).
- 29 ≡ 1 (mod 7). Four rounds (2 processes). Level 4 (2 weeks).

**The cycle of the month:**

The calendar calls this "30 days".
But mathematically, 30 (2 × 15)
is only half of the process.
To get the real, complete cycle of the month, we have to do the process once more: 2 × 30 = 60.

- 30 = 2 weeks (2 processes, 4 moments). This is half of the month cycle.
- 60 = 4 weeks (4 processes, 8 moments). This is the complete month cycle.

Just as on the clock we do not say
"it goes up to 12", but we use up to 24
to describe the day.

So for the month we do not say "it is only 30", but we use 60 (4 weeks)
to show the complete cycle.

**The double cycle of the year (6 and 12):**

Just as 12 is double 6,
the year (12 months) is the double cycle of half a year (6 months).

- The 6 is linked to the week, because 7 ≡ 1 (mod 6).
- The year is therefore a beautiful spiral of 2 × 6 months.

**The spiral in the rest of the chain:**

The clock:

- 13 ≡ 1 (mod 12). One round. Half a day.
- 25 ≡ 1 (mod 12). Two rounds. A whole day.
- 37 ≡ 1 (mod 12). Three rounds. A day and a half.

The digital root:

- 10 ≡ 1 (mod 9). One round. (In theory)
- 19 ≡ 1 (mod 9). Two rounds. (Actual)
- 28 ≡ 1 (mod 9). Three rounds. (Etc.)

The week as a number:

- 4 ≡ 1 (mod 3). One round.
- 7 ≡ 1 (mod 3). Two rounds.
- 10 ≡ 1 (mod 3). Three rounds.

The minute:

- 61 ≡ 1 (mod 60). One round: the half hour.
- 121 ≡ 1 (mod 60). Two rounds: the whole hour.
- 181 ≡ 1 (mod 60). Three rounds: an hour and a half.

The position is always 1.
The level rises.
That is the spiral.

## Step 6d – The leap year and the pattern q + 1 = p

There is another pattern.
Look at the minute and the hour.

- 60 × 60 = 3,600. That is 1 hour.
- 61 × 60 = 3,660. That is 1 hour + 1 minute.

So 61 = 60 + 1. And 61 ≡ 1 (mod 60).

And look at the year, on the scale of days.

- 360 days = 1 year.
- 366 days = 1 leap year.

And 366 = 6 × 61. And 61 = 60 + 1.

So the leap year is the year plus 6 days.
And 6 = 7 – 1.

So the pattern q + 1 = p returns at every scale:

- 60 + 1 = 61. And 61 ≡ 1 (mod 60).
- 6 × 61 = 366. And 366 = 360 + 6.
- 7 – 1 = 6. And 7 ≡ 1 (mod 6).

## Step 6e – Every cycle has its own modulus, but the form is general

Every cycle has its own modulus:

- The clock face: mod 12.
- The second: mod 60.
- The minute: mod 60.
- The hour (of the day): mod 24.
- The day of the month: mod 30.
- The month (of the year): mod 12.
- The weekday: mod 7.
- The digital root: mod 9.
- The week as a number: mod 3.

But the form is the same for all of them:

p ≡ 1 (mod q)

And the theoretical form is:

q + 1 = p

And the actual form is:

2q + 1 = p

So there is one general form, and every cycle is an instance of it.

And every cycle has its own theoretical and actual form:

- Theoretical: p = q + 1, m = 1.
- Actual: p = 2q + 1, m = 2.

So there is both a general form
and a modulus of its own for each cycle.

## Step 6f – The three returns everywhere

There is something else remarkable.
The three returns everywhere.
Let us go through all the units and see where the three sits.

- The millisecond: the smallest unit.
- The second: 1,000 milliseconds. And 1,000 ≡ 1 (mod 3).
- The minute: 60 seconds. And 60 ≡ 0 (mod 3).

- The hour: 60 minutes. And 60 ≡ 0 (mod 3).
- The day: 24 hours. And 24 ≡ 0 (mod 3).
- The week: 7 days. And 7 ≡ 1 (mod 3).
- The month: 30 days. And 30 ≡ 0 (mod 3).
- The year: 12 months. And 12 ≡ 0 (mod 3).
- The decade: 10 years. And 10 ≡ 1 (mod 3).
- The century: 10 decades. And 10 ≡ 1 (mod 3).
- The millennium: 10 centuries. And 10 ≡ 1 (mod 3).
- The digital root: 9. And 9 ≡ 0 (mod 3).

So the three returns in:

- 1,000 ≡ 1 (mod 3).
- 60 ≡ 0 (mod 3).
- 24 ≡ 0 (mod 3).
- 7 ≡ 1 (mod 3).
- 30 ≡ 0 (mod 3).
- 12 ≡ 0 (mod 3).
- 10 ≡ 1 (mod 3).
- 9 ≡ 0 (mod 3).

Check: all remainders are 0 or 1. None of the bases in the list is ≡ 2 (mod 3).

And 9 ≡ 0 (mod 3) is important.
Because 9 is the modulus of the digital root.
And 9 is divisible by 3. And 9 ≡ 0 (mod 3).

And 12 has digital root 3.
Because 1 + 2 = 3. So 12 ≡ 3 (mod 9).

So 12 is the digital root 3. And 12 ≡ 0 (mod 3).

Without beating around the bush: we do not keep counting up from 9, although that is of course possible, as we are used to. 12 stays the same, but connected to the three.
So the three returns everywhere. In time, in the days, in the digital root.

## Step 7 – The same 12 in two places

Now we go to the calendar. Look at the year. How many months does a year have? Twelve.

What you see here: the same number 12 that was just on the clock face is now on the calendar. There are twelve hours in a half day, and twelve months in a year.

But mathematically
they are not two separate twelves.
It is the same 12
at two different scales:

- On the scale of the day: 24 hours = 2 · 12. Two rounds of twelve hours.
- On the scale of the year: 12 months = 1 year. One round of twelve months.

Both are the same cycle,
only at a different scale.
What you see here is that the clock and the calendar
are not two separate systems.
They are one system with the same structure
at different scales.

## Step 7b – The transition from clock to calendar

This is the moment to pause

at something special.
The clock and the calendar share the same q.
Both have q = 12.

- The clock has 12 positions on the clock face.
- The calendar has 12 months in a year.

That means the form p ≡ 1 (mod q)
has the same q for both.
And so also the same p-values:

- Theoretical (m = 1): p = 13. On the clock: 13 hours = 1 hour. On the calendar: 13 months = month 1.
- Actual (m = 2): p = 25. On the clock: 25 hours = 1 hour. On the calendar: 25 months = month 1.
- At m = 3: p = 37. On the clock: 37 hours = 1 hour. On the calendar: 37 months = month 1.
- At m = 4: p = 49. On the clock: 49 hours = 1 hour. On the calendar: 49 months = month 1.
- At m = 5: p = 61. On the clock: 61 hours = 1 hour. On the calendar: 61 months = month 1.
- At m = 6: p = 73. On the clock: 73 hours = 1 hour. On the calendar: 73 months = month 1.
- At m = 7: p = 85. On the clock: 85 hours = 1 hour. On the calendar: 85 months = month 1.
- At m = 8: p = 97. On the clock: 97 hours = 1 hour. On the calendar: 97 months = month 1.

The clock and the calendar are therefore the same parameter choice at two different scales. They have the same q = 12,
the same p = 25 for m = 2,
and the same structure.
That is what makes the transition from clock to calendar so special.
Because it is not a transition to a new system. It is the same system,
only at a larger/different scale.

## Step 8 – The month as 2 × 15 and the week modularly linked

Now we come to something else important.
The week is modularly linked to the month.

The week is 15 ≡ 1 (mod 7), m = 2.

The month is 30 = 2 × 15.

So the month is twice the week's p-value (two processes).

And 15 ≡ 1 (mod 7). So 15 is the actual form of the week (1 week with two moments).
So the month (30) is 2 × 15. And 30 ≡ 2 (mod 7). Because 30 = 4 × 7 + 2.
(Just as an hour has 4 quarters.)

In days, that is 4 weeks of 7 days and 2 days left over.
Those 2 days are the four leftovers (4 half days)
from "Not a sum, but a spiral".

So the month is m = 2 for the month itself,
and the month is 2 (mod 7) for the week.
Note: here we get, for the first time, a remainder larger than 1 for a unit of the chain.
With mod 3 the remainders were only 0 or 1.

And so the month is here
modularly linked to the week.

And you can carry this on:

- The year: 360 = 12 × 30 = 24 × 15. And 360 ≡ 3 (mod 7).
- The decade: 3,600 = 240 × 15. And 3,600 ≡ 2 (mod 7).
- The century: 36,000 = 2,400 × 15. And 36,000 ≡ 6 (mod 7).
- The millennium: 360,000 = 24,000 × 15. And 360,000 ≡ 4 (mod 7).

In this way the whole chain is modularly linked
to the week.

And the week is no longer separate.
The week is a parallel cycle that is modularly linked to the month and the rest of the chain.

## Step 9 – When the clock closes

Our clock consists of two cycles running side by side:

- The calendar is 1,000 years of 360 days, so 360,000 days.
- The week is 7 days.

They come together again at the same starting point
after the least common multiple (lcm)
of both lengths:

$360{,}000 = 2^6 \cdot 3^2 \cdot 5^4$
7 does not occur in it.

lcm(360,000; 7) = 7 · 360,000 = 2,520,000 days

2,520,000 days is 2,520,000 ÷ 360 = 7,000 years.

In milliseconds that is 2,520,000 · 86,400,000 = 217,728,000,000,000.
That is the same number as the product of all the bases in the list.

That the product and the cycle coincide here is because 7 shares no prime factor with 360,000
(which is built from 2, 3 and 5).
The lcm of two numbers without a common divisor is their product.
What you see here is that the full clock cycle lasts 2,520,000 days, or 7,000 years.
The number comes from two cycles
waiting for each other.

## Step 10 – The 1000-year cycle and the 7000-year cycle

The clock can close in different ways. That depends on which week length
you take, and which calendar.

**The 7000-year cycle**
with the week of 7 days
and the calendar of 360,000 days:
lcm(360,000; 7) = 2,520,000 days = 7,000 years.

**The 1000-year cycle**
with the week of 6 days
and the calendar of 360,000 days:
lcm(360,000; 6) = 360,000 days = 1,000 years.

But with a calendar of 360 days
and a week of 7 days:
lcm(360; 7) = 2,520 days = 7 years.

And with a calendar of 360 days and a week of 6 days:
lcm(360; 6) = 360 days = 1 year.

So the 1000-year cycle arises
when you take the week of 6 days
at the scale of millennia.

And the 7000-year cycle arises
when you take the week of 7 days
at the scale of millennia.

Both are valid.
It depends on which week you choose.

And you can carry this on up to 10,000.

10,000 years = 10,000 × 360 = 3,600,000 days.
And 3,600,000 ≡ 5 (mod 7).
Because 3,600,000 = 514,285 × 7 + 5.

So the series runs from ms to 10,000 years, and the modular link with the week continues to hold.
And note: additions and cycles
are not the same thing.
A cycle is finite and adding up is not.

## Step 11 – What 2,520 is

You have seen it pass by several times.
But what does it stand for?

2,520 is the smallest number that is divisible
by all the numbers from 1 to 10.

In other words, the lcm of 1 to 10.

There is a second way
to see the same number.
A year of 360 days is divisible by
1, 2, 3, 4, 5, 6, 8, 9 and 10.
Which one is missing? The 7.

Add it, and you get:
360 · 7 = 2,520 = lcm(1, 2, …, 10)

That is why the week of 7 days
is the only choice between 1 and 10
that lengthens the cycle.

Every other week length already fits into 360.
2,520 is the smallest number
that 360 and 7 reach together.
And 2,520,000 is 2,520 · 1,000.

## Step 12 – And one more observation

Now we look at a different system,
one that has nothing to do with time:

N = 19A + 9B

For N = 2,520:
2,520 = 19 · 90 + 9 · 90

So A = B = 90.
The sum 19 + 9 is 28,
and 2,520 = 28 × 90.

That is why every multiple of 28
has a symmetric pair A = B.
For 2,520 that is A = B = 90.

The digital root of 90 is 9,
and that of 2,520 is also 9.
That is because in this system, for every representation, dr(A) = dr(N),
since 19 ≡ 1 (mod 9).

What you see here:
2,520 is divisible by 28 = 19 + 9,
so it has the symmetric representation
A = B = 90.

The clock and the (19, 9) system arrive here
at the same number.

More examples in the (19, 9) system:

- Other representations of 2,520: 19 · 99 + 9 · 71 = 1,881 + 639 = 2,520, and 19 · 81 + 9 · 109 = 1,539 + 981 = 2,520. Here A is 99, 90 and 81. The digital root of each of those three is 9, the same as that of 2,520.
- Smaller symmetric pairs: 28 = 19 · 1 + 9 · 1, 56 = 19 · 2 + 9 · 2, 84 = 19 · 3 + 9 · 3, 112 = 19 · 4 + 9 · 4.
- 2,521 in the (19, 9) system: 2,521 = 19 · 1 + 9 · 278. The digital root of 2,521 is 1 (2 + 5 + 2 + 1 = 10, and 1 + 0 = 1), and that of A = 1 is also 1.

The first A of 2,520 in N = 19A + 9B is A = 9.
Because the digital root of 2,520 is 9 (2 + 5 + 2 + 0 = 9),
and A has the same digital root.

Then 19 · 9 = 171.
What remains is 2,520 − 171 = 2,349,
and 2,349 = 9 · 261. So B = 261.

Check: 19 · 9 + 9 · 261 = 171 + 2,349 = 2,520.

## Step 12b – Synchronization with 2,520

2,520 is the tool here for testing
whether a cycle fits with the rest of the system. A modulus q synchronizes with 2,520
if both q and 2q
(the two rounds of the actual form)
are divisors of 2,520.
That is the same as: q is a divisor of 1,260.

The test applies to the modulus q, not to p.
So p = 25 is not a divisor of 2,520
(2,520 = 100 · 25 + 20), but q = 12 is,
and that is the modulus that 25 belongs to.

Why this works: $2{,}520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$.
Because 2q must also fit,
q may have one factor 2 less:
so q must fit within $2^2 \cdot 3^2 \cdot 5 \cdot 7 = 1{,}260$.

**Synchronous:**

- q = 12 (clock, p = 25): 2,520 ÷ 12 = 210 and 2,520 ÷ 24 = 105.
- q = 7 (week, p = 15): 2,520 ÷ 7 = 360 and 2,520 ÷ 14 = 180.
- q = 9 (digital root, p = 19): 2,520 ÷ 9 = 280 and 2,520 ÷ 18 = 140.
- q = 3 (week as a number, p = 7): 2,520 ÷ 3 = 840 and 2,520 ÷ 6 = 420.
- q = 18 (p = 37): 2,520 ÷ 18 = 140 and 2,520 ÷ 36 = 70.
- q = 30 (month, p = 61): 2,520 ÷ 30 = 84 and 2,520 ÷ 60 = 42.
- q = 60 (minute, p = 121): 2,520 ÷ 60 = 42 and 2,520 ÷ 120 = 21.

## Step 12c – Two forms compared: 24 ≡ 1 (mod 23) and 25 ≡ 1 (mod 12)

Both statements are correct as sums.
Side by side:

| | **24 ≡ 1 (mod 23)** | **25 ≡ 1 (mod 12)** |
|---|---|---|
| p | 24 | 25 |
| q | 23 | 12 |
| m = (p − 1) / q | 1 | 2 |
| form | theoretical (q + 1 = p) | actual (2q + 1 = p) |
| positions | 23 | 12 |
| 2,520 ÷ q | 109.56… | 210 |
| 2,520 ÷ 2q | 54.78… | 105 |

What you see here:

- **Positions.** With mod 23 there are 23 positions. Hour 24 then falls on hour 1, while the day has 24 different hours. With mod 12 there are 12 positions. And 24 hours are exactly two rounds of 12.
- **Divisors.** 24 − 1 = 23 is a prime number, so q can only be 23 (or 1). 25 − 1 = 24 has eight divisors: 1, 2, 3, 4, 6, 8, 12 and 24. The clock face (q = 12) is among them.
- **2,520.** For q = 12, 2,520 fits exactly 210 times, and 2q = 24 fits 105 times. For q = 23 no whole number comes out.
- **Next p-values.** For q = 23 these are 24, 47, 70, 93. For q = 12 these are 13, 25, 37, 49.

So 24 ≡ 1 (mod 23) is a valid form.
But 25 ≡ 1 (mod 12) fits the clock face and 2,520.

## Step 13 – The general form

Now we look at the structure of the two systems. Not at the numbers,
but at the form.

The (19, 9) system:
19 ≡ 1 (mod 9), m = (19 − 1) / 9 = 2

The clock:
25 ≡ 1 (mod 12), m = (25 − 1) / 12 = 2

The form is identical. In both cases there is a number p that is one more than a multiple of another number q, and the ratio m equals 2.

We can write that in general as:

p ≡ 1 (mod q)

and the ratio:

m = (p − 1) / q

This is the general form.
What you see here is that the clock
and the (19, 9) system are two instances
of that same form.

(25, 12) and (19, 9)
are two parameter choices
within the same general form.
For the clock that is (p, q) = (25, 12).
For the number system it is (p, q) = (19, 9). Both have m = 2.
And both follow the same rules.

## Step 14 – The family of parameter choices

The general form p ≡ 1 (mod q) is not limited to two cases.

For every cycle q you can take p = 2q + 1,
and then m = 2.

- The clock and the months: p = 25, q = 12, check 25 ≡ 1 (mod 12), m = 2.
- The week: p = 15, q = 7, check 15 ≡ 1 (mod 7), m = 2.
- The week as a number: p = 7, q = 3, check 7 ≡ 1 (mod 3), m = 2.
- The number system: p = 19, q = 9, check 19 ≡ 1 (mod 9), m = 2.
- Another example from the number system: p = 37, q = 18, check 37 ≡ 1 (mod 18), m = 2.
- Another example: p = 199, q = 99, check 199 ≡ 1 (mod 99), m = 2.
- Another example: p = 9, q = 4, check 9 ≡ 1 (mod 4), m = 2.
- Another example: p = 11, q = 5, check 11 ≡ 1 (mod 5), m = 2.
- Another example: p = 13, q = 6, check 13 ≡ 1 (mod 6), m = 2.
- Another example: p = 21, q = 10, check 21 ≡ 1 (mod 10), m = 2.
- Another example: p = 29, q = 14, check 29 ≡ 1 (mod 14), m = 2.
- Another example: p = 31, q = 15, check 31 ≡ 1 (mod 15), m = 2.
- Another example: p = 41, q = 20, check 41 ≡ 1 (mod 20), m = 2.
- Another example: p = 61, q = 30 (the month), check 61 ≡ 1 (mod 30), m = 2.
- Another example: p = 121, q = 60 (the minute), check 121 ≡ 1 (mod 60), m = 2.

In each of these systems you have the same kind of cycle and the same kind of representations.
The numbers are different and the form is the same.

## Step 15 – The p-values up to 90 and 100

Now we look at all the p-values up to 90
and up to 100.
For each q you get a series p = m · q + 1.

**For q = 12 (clock and calendar):**

- Theoretical m = 1: p = 13.
- Actual m = 2: p = 25.
- At m = 3: p = 37.
- At m = 4: p = 49.
- At m = 5: p = 61.
- At m = 6: p = 73.
- At m = 7: p = 85.
- At m = 8: p = 97.

The values up to 90 are 13, 25, 37, 49, 61, 73, 85. Up to 100, 97 is added.

**For q = 7 (week):**

- Theoretical (m = 1): p = 8.
- Actual (m = 2): p = 15.
- At m = 3: p = 22.
- At m = 4: p = 29.
- At m = 5: p = 36.
- At m = 6: p = 43.
- At m = 7: p = 50.
- At m = 8: p = 57.
- At m = 9: p = 64.
- At m = 10: p = 71.
- At m = 11: p = 78.
- At m = 12: p = 85.
- At m = 13: p = 92.
- At m = 14: p = 99.

The values up to 90 are 8, 15, 22, 29, 36, 43, 50, 57, 64, 71, 78, 85.
Up to 100, 92 and 99 are added.

**For q = 3 (week as a number):**

- Theoretical (m = 1): p = 4.
- Actual (m = 2): p = 7.
- At m = 3: p = 10.
- At m = 4: p = 13.
- At m = 5: p = 16.
- At m = 6: p = 19.
- At m = 7: p = 22.
- At m = 8: p = 25.
- At m = 9: p = 28.
- At m = 10: p = 31.

The values up to 90 are 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40, 43, 46, 49, 52, 55, 58, 61, 64, 67, 70, 73, 76, 79, 82, 85, 88.
Up to 100, 91, 94, 97 and 100 are added.

**What the tables tell you:**

- The p-values with m = 2 are: 25, 15, 7. Those are the central choices of this document.
- Each p-value with m = 2 is the smallest p that gives m = 2 for that q.
- The p-values with m = 1 are simpler to understand.
- The p-values with m = 3, 4, 5, ... give three, four, five rounds. The higher m, the more rounds.

The limit of 90 comes from the number system
(A = B = 90 for N = 2,520).

The limit of 100 comes from the metric system: a euro has 100 cents,
a quarter-euro coin is 25 cents.
Both limits are useful.

## Step 16 – The role of the 12

Let us look at one more thing:
the role of the 12.
The clock uses the 12 in two places:

- **The hour hand** – There are 12 hours in a half day. After 12 hours the hand is back at its starting point.
- **The month** – There are 12 months in a year. After 12 months the year begins again.

In both places the 12 is the same:
a complete cycle.

The 12 on the clock face is therefore connected to
the day. The 12 on the calendar is connected to the year. And the day and the year are connected by the bases 24 and $30 \cdot 12 = 360$:

1 day = 24 hours • 1 year = 360 days

What you see here: the 12 is the link between
the small scale (the day)
and the large scale (the year).
It occurs in both places,
and it connects them with each other.

## Step 17 – What happens if you change the week

So far we have looked at the clock
as it is. But now we change one number

and see what happens.

The week has 7 days. What if we take 6 days? The calendar is 360,000 days.
A week of 6 days fits into it exactly,
because 360,000 ÷ 6 = 60,000.

The lcm is then:
lcm(360,000; 6) = 360,000 days = 1,000 years
The 6-day week adds nothing:
the clock already closes after the calendar itself.

The 7-day week does:
lcm(360,000; 7) = 2,520,000 days = 7,000 years

You can also take them together:
lcm(360,000; 6; 7) = 2,520,000 days = 7,000 years

The 6 changes nothing, because it is already in the 360. A week of 6 days fits 5 times into a month of 30 days. A week of 7 days does not fit into it.

**Overview of week lengths:**

- Week length 1: closing after 1,000 years.
- Week length 2: closing after 1,000 years.
- Week length 3: closing after 1,000 years.
- Week length 4: closing after 1,000 years.
- Week length 5: closing after 1,000 years.
- Week length 6: closing after 1,000 years.
- Week length 7: closing after 7,000 years.
- Week length 8: closing after 1,000 years.
- Week length 9: closing after 1,000 years.
- Week length 10: closing after 1,000 years.

What you see here: the week of 7 days is the only week length between 1 and 10 that does not fit into 360. As a result the week keeps drifting out of step with the calendar,
and that is why the clock closes only after 7 times
the calendar.

## Step 17b – When it closes: 1,000, 3,000, 6,000, 7,000, 9,000 and 10,000 years

By cycle I mean here:
the moment at which it closes.
The date and the weekday then both stand at their starting point again.
That is the first number of days that is divisible by the calendar cycle
(360,000 days, 1,000 years)
and by the week (7 days).

| Years | Days | Remainder ÷ 7 | Does it close? | Days ÷ 2,520 |
|---|---|---|---|---|
| 1,000 | 360,000 | 4 | no, the week is not yet at 1 | 142.86… |
| 3,000 | 1,080,000 | 5 | no | 428.57… |
| 6,000 | 2,160,000 | 3 | no | 857.14… |
| 7,000 | 2,520,000 | 0 | **yes** | 1,000 |
| 9,000 | 3,240,000 | 1 | no | 1,285.71… |
| 10,000 | 3,600,000 | 5 | no | 1,428.57… |

Only 7,000 years closes.
And only 7,000 years is a whole multiple of 2,520 days (exactly 1,000 times).
That is the synchronization with 2,520.

**Why 3,000, 6,000 and 9,000 do not close**

You might think: 3 × 1,000 = 3,000, 6 × 1,000 = 6,000
and 9 × 1,000 = 9,000,
just like 7 × 1,000 = 7,000.
That only works if the week length shares no prime factor with 360,000 = $2^6 \cdot 3^2 \cdot 5^4$.
The 7 shares none.
The 3 and the 9 share the 3.
The 6 shares the 2 and the 3.
That is why the clock with those weeks already closes after 1,000 years, and no 3,000, 6,000 or 9,000 comes out.

At the small scale the closing is
2,520 days = 7 years.
That is the same number 2,520,
now without the 1,000 years of the millennium.

## Step 17c – The Gregorian cycle

For comparison with the 360-day calendar used here. The Gregorian rule is: a year has 365 days, and 366 days if it is divisible by 4, except century years that are not divisible by 400.

- In 400 years there are 100 – 4 + 1 = 97 leap years.
- Days in 400 years: 400 · 365 + 97 = 146,097.
- 146,097 ÷ 7 = 20,871, so exactly a whole number of weeks.

The Gregorian calendar therefore closes after 400 years: date and weekday then both stand
at their starting point again.
The average year is 146,097 ÷ 400 = 365.2425 days.

The Gregorian cycle closes because the leap rule makes the 400 years a whole number of weeks.
The 360-day calendar closes because 7 is not a divisor of 360,000 and the week therefore multiplies the cycle exactly by 7.

Note: the leap year here is a different leap year than in step 6d.
There, 366 is the year of 360 days plus 6 days.
Here, 366 is the year of 365 days plus 1 day.

## Step 17d – The full series from millisecond to 10,000 years

All units from small to large, with the duration, the remainder on division by 7 days
(the link with the week) and whether the duration is a whole multiple of 2,520 days.

| Unit | Duration | Remainder ÷ 7 days | Multiple of 2,520 days |
|---|---|---|---|
| millisecond | 1 ms | – | no |
| second | 1,000 ms | – | no |
| minute | 60,000 ms | – | no |
| hour | 3,600,000 ms | – | no |
| day | 1 day | 1 | no |
| week | 7 days | 0 | no |
| month | 30 days | 2 | no |
| year | 360 days | 3 | no |
| 7 years | 2,520 days | 0 | yes, 1 × |
| decade | 3,600 days | 2 | no |
| century | 36,000 days | 6 | no |
| millennium | 360,000 days | 4 | no |
| 3,000 years | 1,080,000 days | 5 | no |
| 6,000 years | 2,160,000 days | 3 | no |
| 7,000 years | 2,520,000 days | 0 | yes, 1,000 × |
| 9,000 years | 3,240,000 days | 1 | no |
| 10,000 years | 3,600,000 days | 5 | no |

What the series shows:

- Below the day a unit is not a whole number of days, so there is no remainder on division by 7 and no link with 2,520.
- Remainder 0 occurs only at the week (7 days), at 7 years (2,520 days) and at 7,000 years (2,520,000 days). Only there is the week back at its starting point.
- For the units of the chain the remainder goes up and down: month 2, year 3, decade 2, century 6, millennium 4. That is the modular link with the week from step 8.
- Only 7 years and 7,000 years are a whole multiple of 2,520 days (1 time and 1,000 times). The 7-year cycle closes at the small scale, the 7,000-year cycle at the scale of millennia.

## Step 18 – What the clock finally tells you

Let us put everything in a row.
What does the clock tell us?

1. The clock is a positional system. It uses several bases: 1,000, 60, 60, 24, 7, 30, 12, 10, 10, 10. Every position has its own weight. Every moment in time is a list of numbers.
2. The clock has a bijection within the cycle. There is a one-to-one mapping between the number N (the number of milliseconds since an agreed starting point) and the clock tuple.
3. The clock has a double cycle. The 12-hour clock face has m = 2: two rounds per day (2 × 12 = 24 hours). The p-value of the day is 25, and 25 ≡ 1 (mod 12).
4. The clock and the calendar share the same 12. The 12 of the hour hand and the 12 of the months are the same number at two scales. They also have the same q = 12 and therefore the same p-values: 13, 25, 37, 49, 61, 73, 85, 97.
5. The clock closes after a specific number of days. With the current bases that is 2,520,000 days, or 7,000 years.
6. 2,520 is the least common multiple of 1 to 10.
7. 2,520 is divisible by 28 = 19 + 9. In the (19, 9) system, 2,520 has the symmetric representation A = B = 90.
8. The month is 2 × 15 = 30. And 15 ≡ 1 (mod 7). So the month is twice the week's p-value. And so the month is modularly linked to the week.
9. The whole chain is modularly linked to the week: month ≡ 2 (mod 7), year ≡ 3 (mod 7), decade ≡ 2 (mod 7),

century ≡ 6 (mod 7), millennium ≡ 4 (mod 7).

10. The three returns everywhere: 1,000 ≡ 1 (mod 3), 60 ≡ 0 (mod 3), 24 ≡ 0 (mod 3), 7 ≡ 1 (mod 3), 30 ≡ 0 (mod 3), 12 ≡ 0 (mod 3), 10 ≡ 1 (mod 3), 9 ≡ 0 (mod 3).
11. The clock is a parameter choice within the same general form. The clock and the (19, 9) system use the form p ≡ 1 (mod q) with m = 2.
12. A modulus q synchronizes with 2,520 if q and 2q are both divisors of 2,520. That holds for q = 3, 7, 9, 12, 18, 30 and 60.
13. Two rounds are always one whole unit, and one round is half of it: the half day, the half week, the half month, the half hour.
14. It is a spiral and not a sum. Every round ends at position 1 of the next level. There is no day 0, month 0 or minute 0.
15. The month is 4 × 15 = 60 half days: 4 weeks (28 days) plus the four leftovers (2 days) is 30 days. Four quarters of an hour are likewise one hour: 4 × (2q + 1) = 8q + 4.
16. Only 7,000 years closes, and after that every multiple of 7,000. The 1,000, 3,000, 6,000, 9,000 and 10,000 years do not close.
17. The Gregorian calendar closes after 400 years (146,097 days, 20,871 weeks).

## Step 19 – What we have done

We started with the question: what is a clock? We looked at what everyone knows: the clock counts in 60 seconds, 60 minutes, 24 hours.
We put the numbers in a row:
1,000, 60, 60, 24, 7, 30, 12, 10, 10, 10.

We wrote the clock down as a positional system with a mixed base. We saw how every position has its own weight, and how the bijection between N and the clock tuple works.

Then we looked deeper:

- The clock face has 12 positions. The day has 24 hours. The ratio is 2.
- The calendar has 12 months. The month has 30 days. The week has 7 days.
- It is a spiral: every round ends at position 1 of the next level.
- The month is 4 × 15 = 60 half days, and that is 4 weeks plus the four leftovers.
- The full cycle lasts 2,520,000 days, or 7,000 years.
- And 2,520 is the least common multiple of 1 to 10.
- And 2,520 is divisible by 28 = 19 + 9.
- And the month is 2 × 15 = 30. And 15 ≡ 1 (mod 7). So the month is modularly linked to the week.
- And the whole chain is modularly linked to the week.
- And the three returns everywhere.
- And the clock is a parameter choice within the same general form.
- And only 7,000 years closes. The Gregorian calendar closes after 400 years.

We also looked at what happens when you change the week. With 6 days nothing changes, because 6 is already in 360. And we saw that the choice of the week is a choice with consequences. Not a choice between right and wrong.

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## The week as a parallel cycle

The week is a unit of 7 days.
But it does not fit neatly into the calendar.
A month has 30 days,
and 30 is not a multiple of 7.
A year has 360 days,
and 360 is not a multiple of 7.
Yet the week is all around us.
That is why it deserves a place of its own.
And that place is: a parallel cycle that is modularly linked to the month and the rest of the chain.

## The week runs parallel to the calendar

Imagine: the calendar counts days in months and years. The week counts days in weeks. They start at the same moment,
but they drift out of step.

- After 7 days the week is complete, but the calendar is still in the middle of the first month.
- After 30 days the month is complete, but the week is in the middle of the fifth week.
- After 360 days the year is complete, but the week is again in the middle of a week.

The week and the calendar come together only when their lengths meet at the same point again. That is after 2,520 days.

## What else the week is

The week has 7 days.
But 7 is also a number that fits the form
$p \equiv 1 \pmod{q}$:

$7 = 2 \times 3 + 1$. So $7 \equiv 1 \pmod{3}$. $m = 2$.

So the week is not only a cycle of 7.
The number 7 is everywhere.

$15 \equiv 1 \pmod{7}$, $m = (15 - 1) / 7 = 2$

That means: two full rounds of 7.
And those two rounds together form 1 week with two different moments.

### The week in hours

$15 \equiv 1 \pmod{7}$ is, in hours, 168 hours.
Because $24 \times 7 = 168$.

The digital root agrees too:
$1 + 6 + 8 = 15$, and $1 + 5 = 6$.
Both 168 and 15 have digital root 6.

Half of 168 is 84.
That is the half week.
$8 + 4 = 12$, and $1 + 2 = 3$.
The digital root of 84 is 3.

Just like 12 hours of the 24 hours,
and 30 minutes of the hour.
Each time the half, and each time digital root 3.

Dividing further: $84 \div 2 = 42$, and $84 \div 7 = 12$.
The half week is therefore 7 times 12 hours.
That is one round of 7 half days.

## The quarter hour and the quarter-euro coin

There is another connection.
Look at the hour and the euro.

An hour has 60 minutes.
A quarter of an hour is 15 minutes ($60 \div 4$).
And 15 is also the p-value of the week: $15 \equiv 1 \pmod{7}$.

A euro has 100 cents.
A quarter of a euro is 25 cents ($100 \div 4$).
And 25 is also the p-value of the clock: $25 \equiv 1 \pmod{12}$.

Both quarters come from the same formula:
X = 8q + 4, with as quarter 2q + 1.

- For 60 (hour): X = 8 · 7 + 4, so q = 7, quarter X ÷ 4 = 15, fits the week.
- For 100 (euro): X = 8 · 12 + 4, so q = 12, quarter X ÷ 4 = 25, fits the clock.
- For 28 (4 weeks): X = 8 · 3 + 4, so q = 3, quarter X ÷ 4 = 7, fits the week as a number.

What you see here: the quarter hour (15)
and the p-value of the week (15)
are the same number.
The quarter of a euro (25) and the p-value of the clock (25) are the same number.
That is because 60 = 8 · 7 + 4 and 100 = 8 · 12 + 4,
so the quarters are 2q + 1 for q = 7 and q = 12.

In 8q + 4 the 8q is four times two rounds,
and the +4 are the four leftovers.
Four quarters of an hour together make an hour again,
and after that you are in the next hour.

What the week tells us:
the week is a cycle of 7 days that runs parallel to the calendar.
It does not fit neatly into the month or the year.
But it does stand in the same form as the clock
and the number system.
And the connection with the quarter hour shows that
the week is not isolated:
it belongs to the same structure that underlies all of timekeeping.

## The closing table

Every choice gives a different closing duration.
The closing is the least common multiple of the week cycle and the calendar.

**Without decade, century and millennium (calendar of 360 days):**

- Week of 6 days: closing after 360 days, that is 1 year.
- Week of 7 days: closing after 2,520 days, that is 7 years.
- 6 and 7 together: closing after 2,520 days, that is 7 years.

**With decade, century and millennium (calendar of 360,000 days):**

- Week of 6 days: closing after 360,000 days, that is 1,000 years.
- Week of 7 days: closing after 2,520,000 days, that is 7,000 years.
- 6 and 7 together: closing after 2,520,000 days, that is 7,000 years.

**What this tells us:**

- Only the 7 lengthens the cycle. It is the only week length up to and including 10 that does not fit into 360.
- The 6-day week brings nothing new. It closes together with the calendar.
- The combination of 6 and 7 equals the 7 alone. Because 6 is already in 360.
- The scale is a matter of application. For everyday timekeeping the 7-year cycle is the most useful, and for a long-term calendar the 7,000-year one.
- The 7-day week gives the richest connection with other number systems. Because 7 · 360 = 2,520 = lcm(1…10).

## The clock and the main theory

The clock is an instance of the main theory
p ≡ 1 (mod q).

As far as I know, this formulation,
in which the clock is described as a parameter choice within this form, has not been written down before.
I cannot rule out that it exists elsewhere. Anyone who knows an earlier source is invited to share it.

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